Absolute egocentric 3D hand pose on HOI4D (157 held-out seqs)

Camera-frame metrics in mm, right hand, 16 joints, true focal, per-sequence aggregation. Abs. MPJPE = absolute joint position error (no root alignment); Local MPJPE = root-relative.

Method	Abs. MPJPE ↓	Abs. wrist ↓	Local MPJPE ↓
Ground-truth hand boxes (matched-input regime)			
WiLoR (off-the-shelf)	240.0	–	30.5
HaMeR (off-the-shelf)	168.3	180.4	36.0
HaMeR, metric-tuned transl. head[a]	39.8	48.6	36.0
HaMeR, decoder fine-tuned[a]	26.5	26.0	17.0
HaMeR, full fine-tune (632M)[a]	21.4	24.1	11.6
HaMeR, full fine-tune + jitter aug[d]	**20.8**	**21.7**	**11.6**
Ours (frozen features + light head)	23.6 ± 0.8	22.7 ± 0.2[b]	17.2 ± 0.4
Ours + box-jitter augmentation[d]	27.3	26.1	17.0
Detector hand boxes (end-to-end regime)			
WiLoR, native full-res pipeline	83.4	–	27.2
HaMeR, native full-res pipeline[c]	88.0	–	30.3
HaMeR, metric-tuned transl. head[a]	68.7	72.2	39.9
HaMeR, full fine-tune (632M), own detector[a]	44.0	43.7	16.1
HaMeR, full fine-tune + jitter aug, own detector[d]	41.6	40.3	14.9
HaMeR, full fine-tune + jitter aug, our detector boxes[d]	**23.4**	**22.3**	**13.2**
Ours (our detector boxes)	49.8	42.9	41.0
Ours + box-jitter augmentation (our detector boxes)[d]	35.6	34.8	26.5
Global / world-space methods (reference only, not the same protocol)			
Hand3R[e]	(42.6)	–	–
HaWoR[e]	(51.8)	–	–
HaPTIC (our run on HOI4D)[f]	n/a	–	25.7

[a] Trained by us on the same 367 HOI4D train sequences with the same absolute 3D keypoint loss as ours (translation head only / MANO decoder / full 632M network, 3–4 epochs). [b] Mean of 3 seeds (22.9, 22.4, 22.8), std 0.2 mm. [c] HaMeR's own detector is build-incompatible; WiLoR detector with HaMeR-native box rescale substituted. [d] Box-jitter augmentation, identical scheme (amplitude 0.2) for ours and the HaMeR control. On *identical* detector boxes (our detector's box set, the same boxes behind the "Ours" rows) the jitter-matched full fine-tune reaches 23.4 mm and degrades less from its GT-box score (+4.4 mm) than our head does (+8.3 mm): an earlier apparent robustness advantage of the frozen head came from comparing across different detector-box sources (HaMeR's own detector produces harder boxes), not from the head itself. The control was retrained for this eval and reproduces the original fine-tune within run-to-run variance (GT-box 19.0 vs 20.8). [e] Numbers as reported in their papers on different datasets and protocols; listed for context only, not directly comparable. [f] HaPTIC's published conversion yields non-metric absolute depth on our data (native-resolution rerun); its local MPJPE is valid.